

Image resampling

Write a function **ImageResample**, which takes as input an [MxN] image matrix **img** that we would like to resample and a [2x1] vector **dim**, which contains the desired resampled image, using the `interp2` and `meshgrid` functions as shown in the lecture. The function outputs **img_resamp**, the resampled version of image with data dimensions **dim**.

(Note that MATLAB's built in function `imresize` is typically used to resample images because it implements anti-aliasing filters when changing the size of an image understanding how to resample an image grid using linear interpolation as we do in this problem is also a useful skill!)

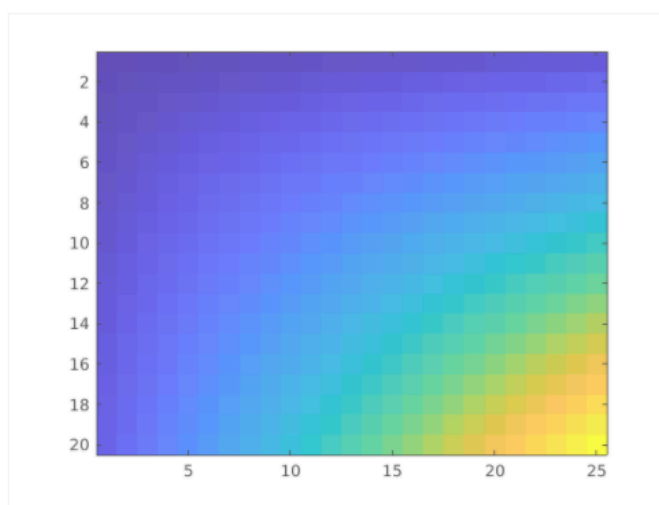
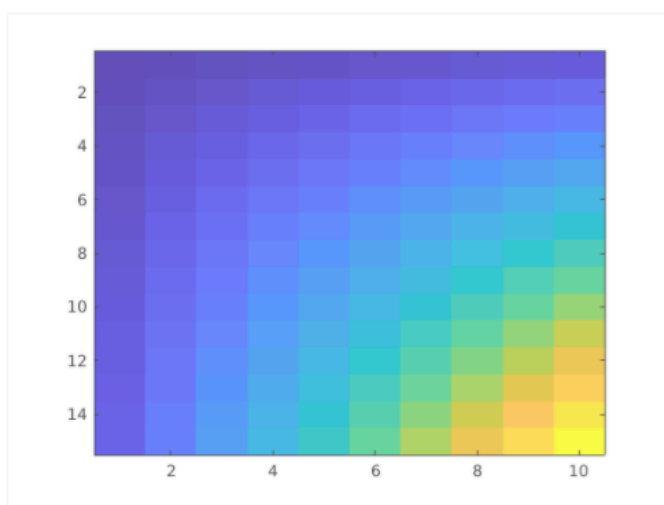
Running the example code below, you should get output identical to this:

ans =

15 10

ans =

20 25



Function ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function img_resample = ImageResample(img, dim)
2 % define r,c,nr,nc
3 [r,c] = size(img);
4 nr = dim(1);
5 nc = dim(2);
6 % use the meshgrid function to determine sampling coordinates
7 % use interp2 to interpolate
8 [C,R] = meshgrid(1:(c-1)/(nc-1):c, 1:(r-1)/(nr-1):r);
9 %[C,R] = meshgrid(1:(nc-1)/(c-1):nc, 1:(nr-1)/(r-1):nr);
10 img_resample = interp2(double(img), C, R, 'linear');
11 end
```

Code to call your function ?

Reset

```
1 [C,R] = meshgrid(1:10,1:15); % inputs are ordered columns, then rows
2 img = uint8(C.*R);
3 size(img) % = 15 x 10
4 dim = [20;25];
5 img_resample = ImageResample(img, dim);
6 size(img_resample) % = 20 x 25
7 imagesc(img);
8 figure
9 imagesc(img_resample)
```

[▶ Run Function](#)

Assessment: All Tests Passed

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✔ Is output correct with random inputs?

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